Surface area

Key idea

Surface area is the total area of all the outside faces of a 3D shape. A good method is to imagine the net, find the area of each face, then add them all together.

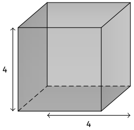
Steps

1. Identify every outside face.
2. Find the area of each face.
3. Multiply pairs of congruent faces when possible.
4. Add the areas and give square units.

Common mistake

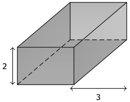
Missing one face. For example, a closed cuboid has 6 faces and a closed cylinder has 2 circles plus 1 curved surface.

Cube



A cube has side length 4 cm.
Each face: $4 \times 4 = 16$ cm^2
6 faces: $6 \times 16 = 96$ cm^2
Surface area = 96 cm^2

Cuboid



A box is 8 cm by 3 cm by 2 cm.
Top & bottom: $2(8 \times 3) = 48$
Front & back: $2(8 \times 2) = 32$
Left & right: $2(3 \times 2) = 12$
Total: $48 + 32 + 12 = 92$ cm^2
Surface area = 92 cm^2